Biochemistry Cumulative Exam

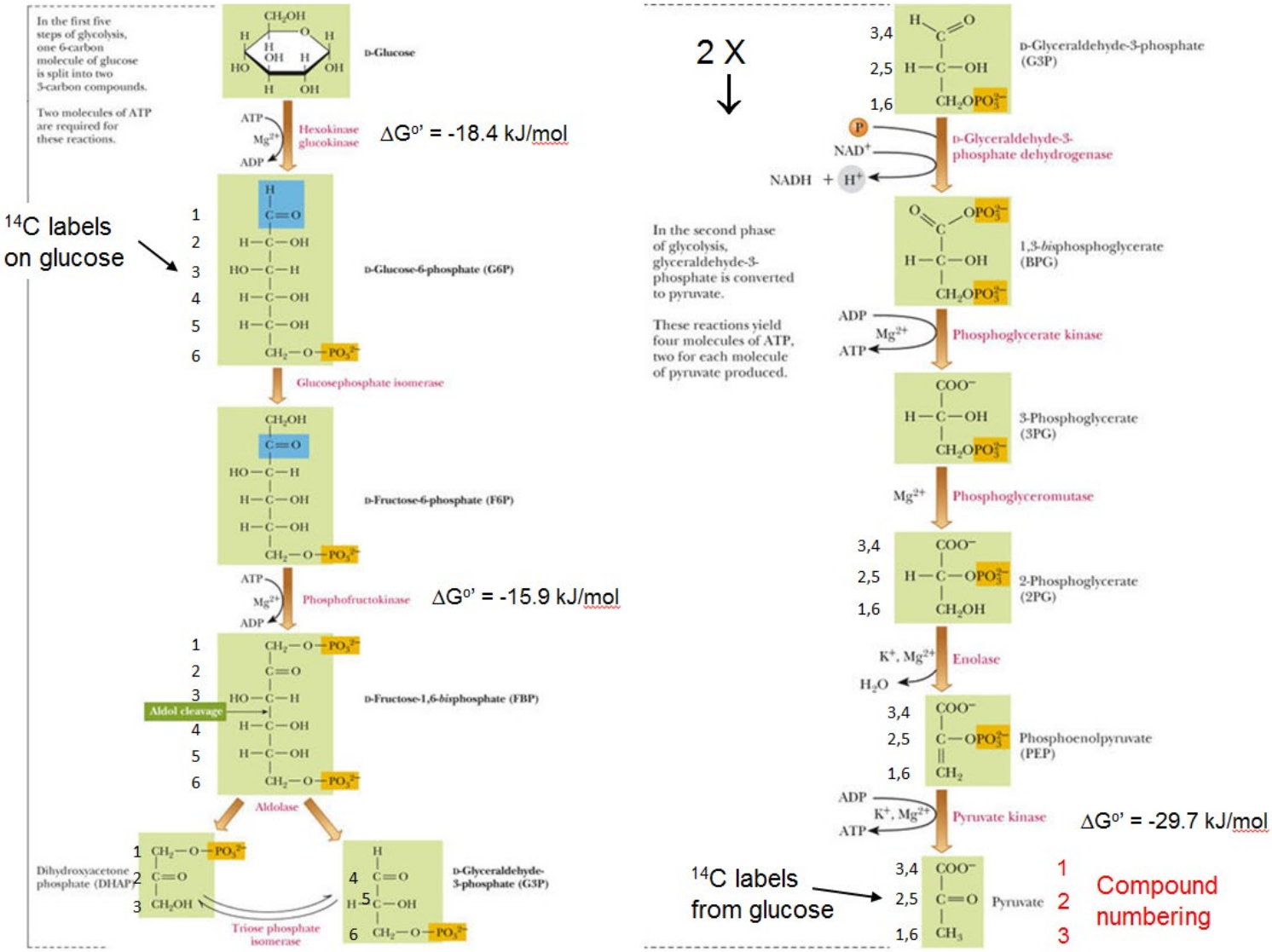
(Supplemental Information Attached)

A mouse is fed a diet including glucose and NH_4^+, but no other carbohydrates, amino acids, citric acid cycle intermediates, or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

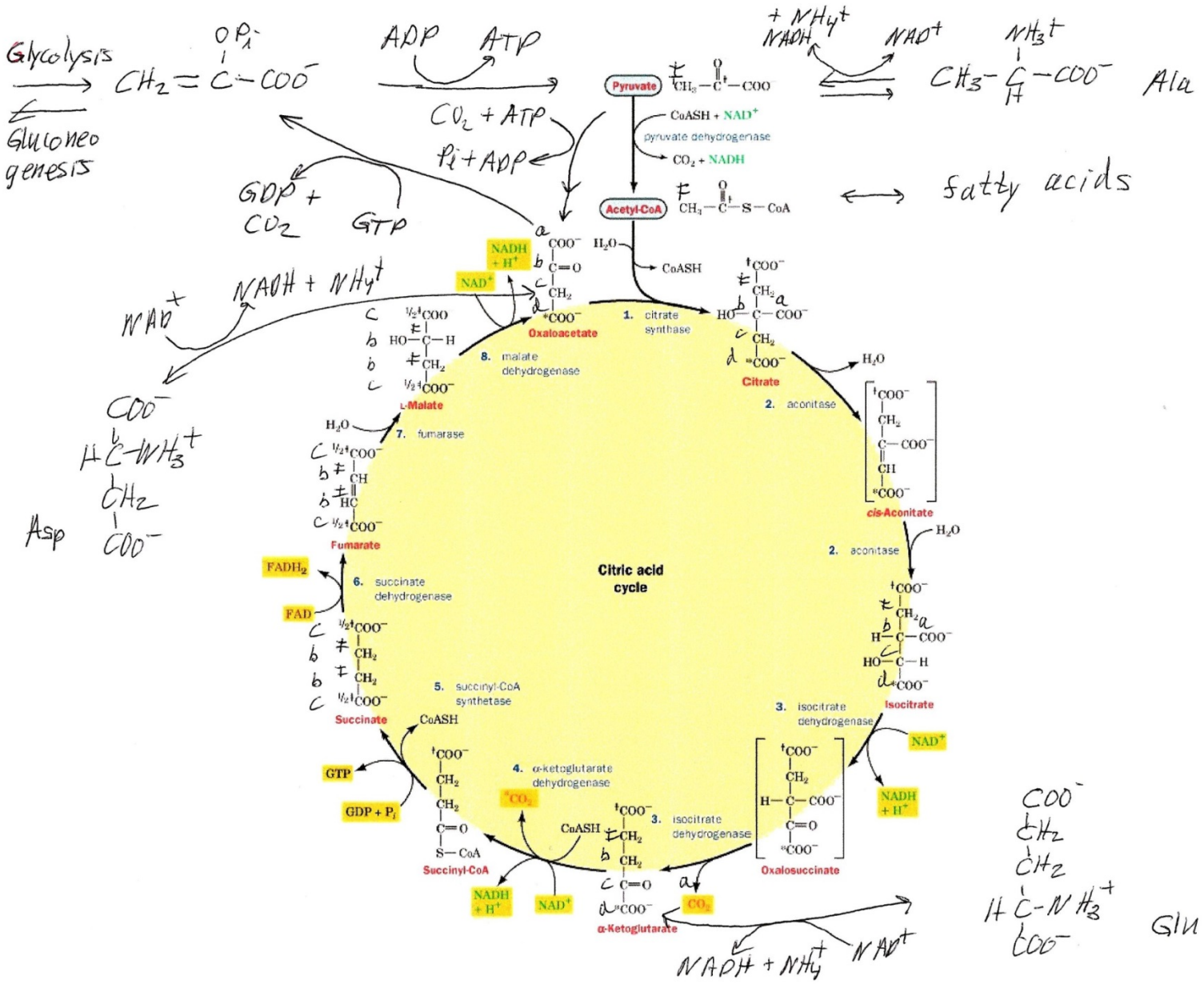
1. (70) Outline a metabolic pathway for the mouse to synthesize glutamate on this diet using glucose and NH_4^+ as the <u>sole source</u> of carbon, nitrogen and energy. How many glucose and NH_4^+ molecules must be used to provide enough carbon atoms, nitrogen, ATP, and NADH molecules to synthesize each molecule of glutamate? Write a complete, balanced net equation for this process giving the numbers of ATP, GTP, NADH, $FADH_2$ and CO_2 generated.

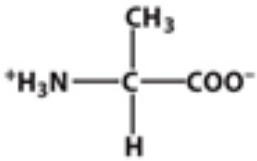
2. (30) If glucose is labeled with ^{14}C at carbon 2, which carbon(s) of glutamate will first become labeled during these reactions.

Glycolysis

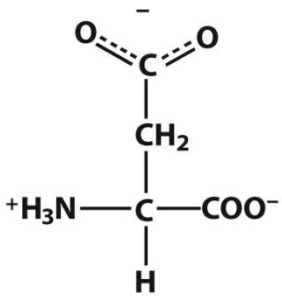

In the first five steps of glycolysis, one 6-carbon molecule of glucose is split into two 3-carbon compounds.
Two molecules of ATP are required for these reactions.
D-Glucose
Hexokinase glucokinase
ΔG°' = -18.4 kJ/mol
14C labels on glucose
D-Glucose-6-phosphate (G6P)
Glucosephosphate isomerase
D-Fructose-6-phosphate (F6P)
Phosphofructokinase
ΔG°' = -15.9 kJ/mol
Aldol cleavage
D-Fructose-1,6-bisphosphate (FBP)
Aldolase
Dihydroxyacetone phosphate (DHAP)
D-Glyceraldehyde-3-phosphate (G3P)
Triose phosphate isomerase
2 X
D-Glyceraldehyde-3-phosphate (G3P)
D-Glyceraldehyde-3-phosphate dehydrogenase
NADH + H+
In the second phase of glycolysis, glyceraldehyde-3-phosphate is converted to pyruvate.
These reactions yield four molecules of ATP, two for each molecule of pyruvate produced.
1,3-bisphosphoglycerate (BPG)
Phosphoglycerate kinase
3-Phosphoglycerate (3PG)
Phosphoglyceromutase
2-Phosphoglycerate (2PG)
Enolase
Phosphoenolpyruvate (PEP)
Pyruvate kinase
ΔG°' = -29.7 kJ/mol
14C labels from glucose
Pyruvate
Compound numbering

Citric Acid cycle

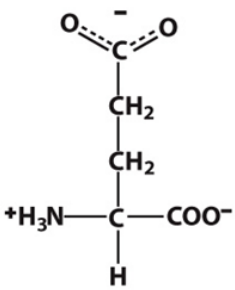




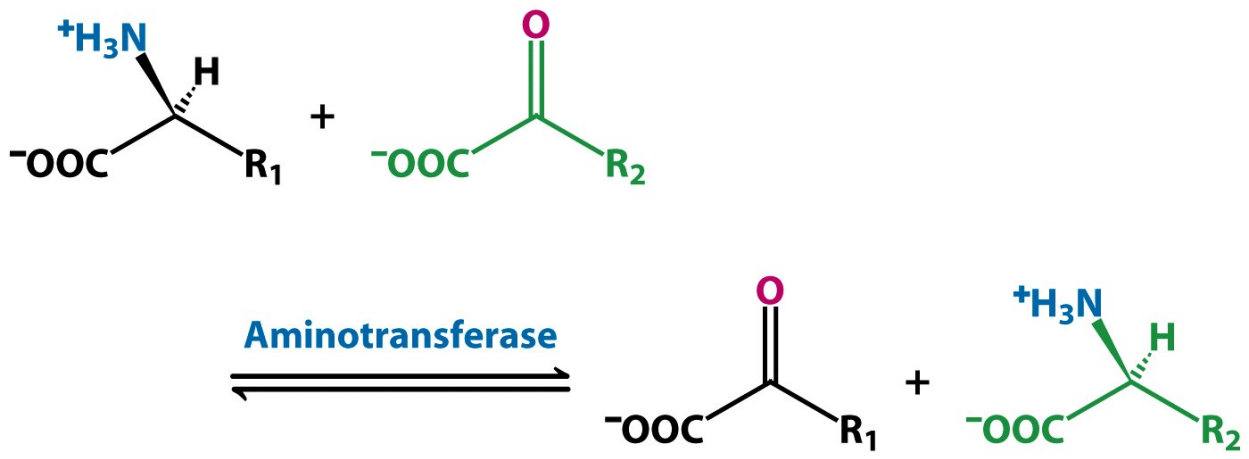
Alanine
(Ala, A)



Aspartate
(Asp, D)



Glutamate
(Glu, E)



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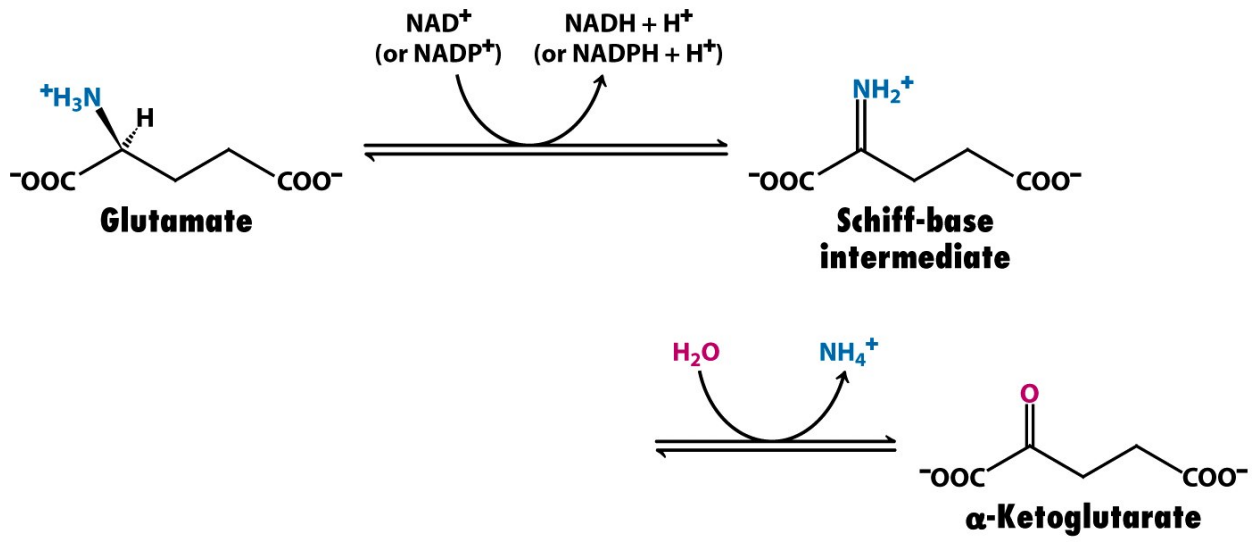

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